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Anterior pelvic tilt increases hamstring strain and is a key factor to target for injury prevention and rehabilitation

Jurdan Mendiguchia[1] | **Mirian Aranzazu Garrues**[2] | **Ernest Schilders**[3,4] | **Gregory D. Myer**[5,6,7,8,9] | **Miki Dalmau-Pastor**[10,11]

[1]Department of Physical Therapy, ZENTRUM Rehab and Performance Center, Barañain, Spain

[2]Physiology Department, Faculty of Medicine and Nursing, University of the Basque Country, Biscay, Spain

[3]Department of Orthopaedic Surgery, Fortius Clinic—FIFA Medical Centre of Excellence, London, UK

[4]Carnegie School of Sport, Leeds Beckett University, Leeds, UK

[5]Emory Sports Performance and Research Center (SPARC), Flowery Branch, Georgia, USA

[6]Emory Sports Medicine Center, Atlanta, Georgia, USA

[7]Department of Orthopaedics, Emory University School of Medicine, Atlanta, Georgia, USA

[8]The Micheli Center for Sports Injury Prevention, Waltham, Massachusetts, USA

[9]Youth Physical Development Centre, Cardiff Metropolitan University, Wales, UK

[10]Human Anatomy and Embryology Unit, Department of Pathology and Experimental Therapeutics, School of Medicine and Health Sciences, University of Barcelona, Barcelona, Spain

[11]MIFAS by GRECMIP (Minimally Invasive Foot and Ankle Society), Merignac, France

Correspondence

Jurdan Mendiguchia, Department of Physical Therapy, ZENTRUM Rehab and Performance Center, Calle B Nave 23 (Polígono Barañain), 31010 Barañain, Spain.
Email: jurdan24@hotmail.com

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Abstract

Purpose: Hamstring muscle strain injury is very common in sports involving high-speed running. Hamstring muscles originate from the ischial tuberosity and thus pelvic position may influence hamstring strain during different sports movements like sprinting, but this has only been evaluated by indirect methods. This study tested the hypothesis that a change in anterior pelvic tilt causes elongation of the overall hamstring complex and disproportionately elongates proximal relative to distal muscle regions.

Methods: Seven fresh-frozen specimens (full lower limb with pelvis and lumbar spine) were used for this in vitro study. Specimens were dissected to enable visualization of the hamstring muscles and then fixed into a custom-made testing bench that allowed controlled movement of the pelvis over a fixed femur and tibia. Nine markers were inserted into the hamstring muscles to allow intra- and intermuscle difference measurements. Then, six different anterior pelvic angles were used to measure the difference in hamstring muscle lengthening through a three-dimensional reconstruction system based on stereoscopic machine vision technology.

Results: An increase in anterior pelvic tilt produced a significant non-uniform increase in tissue elongation in all regions of the three hamstring muscles (semitendinosus, semimembranosus [SMB] and biceps femoris long head), which was greater in the proximal (>1 cm every 5°) compared to the distal region (≈0.4 cm every 5°). At the proximal hamstring region, SMB showed significantly greater length changes compared to conjoint tendons with nonstatistically significant elongation differences between muscles at the distal region.

Conclusion: Considering the results of the study, the pelvis segment will likely play a fundamental role as a strain regulator of hamstring muscles. These results will have an impact on injury rehabilitation and prevention processes of hamstring injuries, as well as optimize future musculoskeletal models and avoid potential underestimation of the hamstring muscle-tendon complex lengthening during high-speed running.

Level of Evidence: N/A.

KEYWORDS

anterior pelvic tilt, hamstring injury, hamstring lengthening, hamstring rehabilitation, hamstring strain, hamstring tendinopathy

Abbreviations: BF, biceps femoris; CJT, conjoint tendon; SMB, semimembranosus; ST, semitendinosus.

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INTRODUCTION

Hamstring strain injury is the most prevalent noncontact injury in high-speed running sports such as Australian football, basketball, baseball, American football, rugby union, soccer and athletic competitions [1–7]. In addition to the high incidence as a primary injury in running and cutting sport athletes, hamstring strain injury has a high risk of recurrence (12%–31%). Cumulatively, hamstring strain injuries can cause significant economic and performance impacts on both the team and the player [8, 9]. Despite the implementation of several strategies to reduce injury risk, without a clear mechanistic understanding of pathomechanics, overall hamstring injury and reinjury rates have not improved over the last three decades [10].

The use of advanced musculoskeletal models and simulation techniques has provided an in-depth understanding of the function of the hamstring muscle-tendon complex during high-speed running, yielding preliminary insights into potential injury mechanisms [11, 12]. Indeed, translational and applied science research shows that hamstring strain, as a result of disproportionate lengthening, is the primary determining factor for muscle damage [13]. Specifically, computer simulation highlights hamstring strain as a primary factor of injury and therefore a key target for hamstring strain injury preventive strategies.

However, controversy remains about in which exact moment of the late swing or stance phase hamstring injury occurs. In these two phases, the hamstring muscles face the highest mechanical strain [14], and in both moments, a large anterior pelvic tilt is present [15]. Theoretically, with a greater anterior pelvic tilt, the ischial tuberosity moves superiorly and posteriorly, potentially resulting in a disproportionate active lengthening and passive tension demand, especially for the proximal region of the posterior thigh musculature [16]. This biomechanical concept might explain in part the association found between anterior pelvic tilt and hamstring strain injury risk in prospective studies evaluating hamstring strain incidence [17, 18].

Currently, the position of the pelvis is not commonly considered to act as a hamstring strain modulator. Prior studies evaluated this relationship via indirect methods such as computer simulations, where complex musculoskeletal models showed that greater trunk flexion and consequent anterior pelvic tilt resulted in greater elongation of the hamstring muscles compared to a more upright trunk position during sprinting [19]. However, current simulation models rely on anatomical studies in cadavers where only changes in the positions of the hip (femur versus fixed pelvis) and knee joints were considered when calculating changes in hamstring length; these studies did not consider the influence and role of movement of the pelvis on the hamstring muscle complex [20].

Yet, elastography identified an increase in passive tension (rigidity) derived from an increase in anterior pelvic tilt, which indirectly reflects the elongation of the hamstring muscles [21]. Furthermore, a change in pelvic position can influence the results of a flexibility test up to 15°, suggesting the importance of assessing the position of the pelvis when determining the strain of the hamstring muscles during sprinting [22]. An increase in anterior pelvic tilt during sprinting in soccer players was associated with hamstring injuries [18], and a multimodal intervention was able to decrease anterior pelvic tilt around 5° during the late swing phase of sprinting (potentially decreasing hamstring strain incidence) [23].

With limited progress in treatment and prevention strategies for hamstring strain injuries, there is a need for investigations to better delineate the relationship between anterior pelvic tilt and hamstring strain. The aim of this study was to directly quantify the effect of the pelvis position on hamstring lengthening in an in-vitro designed cadaver study. It is hypothesized that an increase in anterior pelvic tilt would cause a measurable elongation of the hamstring complex and would disproportionately elongate the proximal relative to distal posterior thigh muscle regions.

MATERIALS AND METHODS

This research was carried out in the dissection room of the University of Barcelona, following ethics permission under IRB number 00003099.

To determine the effect of pelvis biomechanics on hamstring elongation, we developed an in vitro study model to systematically evaluate the effect of differential anterior pelvic tilt positions concomitant with precise measurement of the deformation patterns of the different hamstring muscle complexes.

Seven fresh-frozen lower limbs were used. Specimens included the full lower limb (foot, ankle, knee, hip), pelvis and lower column, sectioned at D12. To test the specimen, a custom-made testing bench was designed in collaboration with an engineer. The bench allowed us to fix the specimen and control the position of the knee and ankle, as well as perform anterior pelvic tilt, as shown in Figure 1.

The steps of the study included dissection, setup and measurements:

Dissection

Specimens were allowed to thaw at room temperature in water for 8 h before dissection. An anatomical plane-per-plane dissection of thawed specimens was performed as follows: (1) A skin window was created to allow visualization of the posterior compartment muscle

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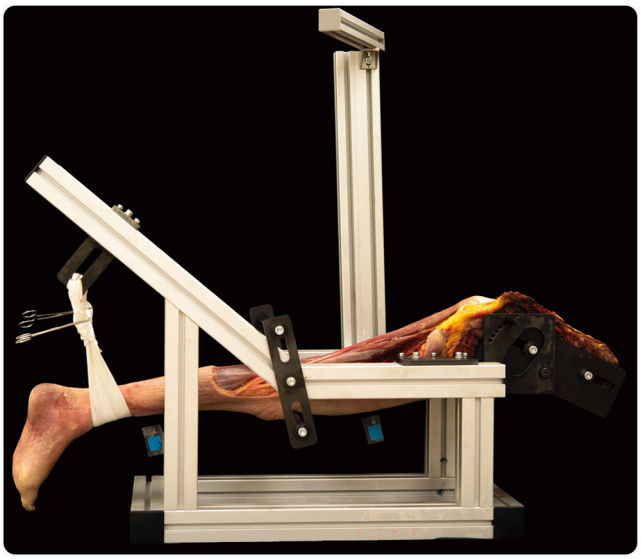

FIGURE 1 Set up of the specimens for the study. After dissecting the posterior side of the thigh, two 22 mm pilot holes were drilled, one between the vertebral bodies of L3–L4 and one in the coxal bone. Then, two metal bars were introduced in the pilot holes to fix the specimen to the custom-made bench built for the study.

fascia. (2) Muscle fascia was dissected to visualize the muscle fibres of the hamstring muscles. (3) To adequately visualize the ischial tuberosity and the sacrotuberosus ligament, the lower part of the gluteus maximus muscle was removed.

Setup

Two 22-mm pilot holes were drilled: one between the vertebral bodies of L3–L4, and one in the coxal bone (2 cm anterior to the iliopubic eminence). Two metal bars were introduced in the pilot holes to fix the specimen to the bench. Nine reference points were marked with needles at: (1) proximal part of the long head of the biceps femoris (BF) and semitendinosus (ST) conjoint tendon (CJT) (1 cm distal to ischial tuberosity); (2) distal part of BF and ST conjoint tendon; (3) midpoint of the BF muscle belly; (4) BF distal tendon (2 cm proximal to fibular head insertion); (5) semimembranosus (SMB) proximal tendon (1 cm distal to ischial tuberosity); (6) SMB distal tendon (2 cm proximal to tibial insertion); (7) ST muscle belly proximal to its raphe; (8) ST muscle belly distal to its raphe and (9) ST distal tendon (2 cm proximal to tibial insertion).

Measurements

A digital inclinometer (Saunders Baseline Digital Inclinometer; Saunders Group Inc.) was positioned on the dorsal part of the third sacral vertebra (pelvic segment) of the specimen. The baseline position was defined as the position where the inclinometer marked 0° (parallel to the ground axis). The hip was flexed 30° and the knee was flexed 20°. These angles were selected to ensure some lengthening of the hamstring muscles, which would facilitate measurement. Then, using the holes located on the rotating plate of the bench, an anterior pelvic tilt was performed on the specimen, from 0° to 25°, with 5° increments (see Figure 2).

Measurements were taken and processed using a three-dimensional (3D) reconstruction system based on stereoscopic machine vision technology. The system included two cameras and one laser (see Figure 3). Cameras were mounted 515 mm from each other, which allowed them to cover a 600 × 400 mm space, and calibrated. From their position, each camera took images of the markers to obtain their precise position in space. Next, the coordinates of markers were manually digitized to calculate the relative position of each marker in three dimensions and check how their position changed along with the angle of the pelvic tilt (see Figure 4).

3D reconstruction of the hamstring muscles was performed using a line laser. The laser was mounted in a linear displacement structure to cover the length of the thigh. As the laser was moved along the surface of the muscles, images were captured by the two mounted cameras and processed by a computer, so that the laser projected onto the muscles was automatically segmented. This information yielded a 3D profile for each pair of images. After the sweep of the laser was finished, all profiles were merged to obtain a cloud of points with the 3D reconstruction of the muscles (see Figure S1). We used 39 images for calibration; the median margin of the measurements after system calibration was 0.02 mm. The maximum registered error was 0.1 mm.

Lengths were measured in mm between the origin and proximal markers (CJT-1, CJT-2, SMB-1), markers in the muscle belly (BF-1, ST-1 and ST-2), and markers in the distal hamstring section (BF-2, SMB-2 and ST-3), and the average change in length was calculated for every 5° of anterior pelvic tilt.

Differences in muscle length (in mm) with 5° in anterior pelvic tilt were also measured within the same section of hamstrings: (1) CJT-1, CJT-2, SMB-1; (2) BF-1, ST-1 and ST-2; and (3) BF-2, SMB-2 and ST-3.

All measurements were made as follows:

(a) Difference between the length in mm measured at 5°, 10°, 15°, 20° and 25° pelvic tilt and the length (mm) at 0° pelvic tilt and calculating the mean difference. That is from 0° to 5°, 0° to 10°, 0° to 15°, 0° to 20° and 0° to 25° (see Figure S2).
(b) Difference in length in mm with 5° increments of anterior pelvic tilt and calculating the mean difference. That is from 0° to 5°, 0° to 10°, 0° to 15°, 0° to 20° and 0° to 25° (see Figure S2).

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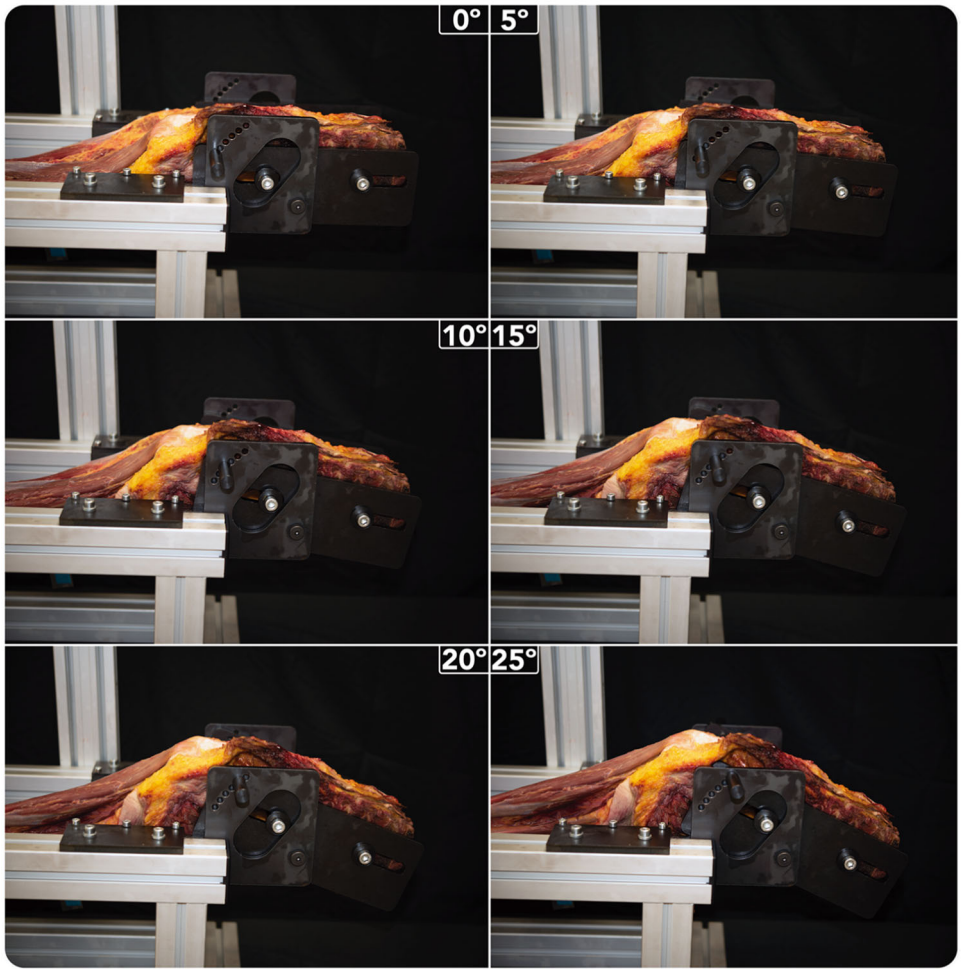

FIGURE 2 Change in anterior pelvic tilt angle of the specimen on the bench. Hamstring muscle lengthening was measured at six angles of anterior pelvic tilt, from 0° to 25°.

Statistical analysis

Paired-sample *T* test was used to compare median length change at hamstring muscle related to anterior pelvic tilt degree. Statistical significance was set at $p < 0.05$.

Three types of measures were made for length increase: (1) between hamstring muscle region changes (proximal CJT-1, CJT-2 and SMB-1, belly BF-1, ST-1 and ST-2, and distal BF-2, SMB-2 and ST-3) with respect to anterior pelvic tilt; (2) within muscle and hamstring region and (3) between muscles at the same hamstring region.

Data analysis was conducted using two types of measurements (a) changes compared to origin position (0° anterior pelvic tilt). That is from 0° to 5°, 0° to 10°, 0° to 15°, 0° to 20° and 0° to 25° anterior pelvic tilt and (b) changes between every 5° anterior pelvic tilt; that is 0–5°, 5–10°, 10–15°, 15–20° and 20–25° anterior pelvic tilt (see Figure S1).

Between muscle regions, comparison analysis was conducted using median changes (a) compared to origin and (b) median changes for every 5° anterior pelvic tilt within and between muscles at proximal region were conducted for CJT-1 and SMB-1 and at distal region for BF-2, SMB-2 and ST-3.

RESULTS

Two types of results are reported: (1) differences between hamstring regions and (2) differences within and between the same region and hamstring muscle at (a) proximal region and (b) distal region.

Differences between hamstring regions (proximal, muscle belly, distal)

Significant changes in muscle length (mm) were observed in the mean median muscle length between the three regions (a) relative to baseline, and (b) for every 5° anterior pelvic tilt increase. These changes were significantly lower at the distal hamstring region compared to the proximal or belly (see Figure 5). No

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differences were found between the proximal hamstring and muscle belly regions at any anterior pelvic tilt angle.

Differences within and between muscle regions

At the proximal hamstring region, significant changes were found for CJT-1 and SMB-1. Both analyses (relative to baseline position and for every 5° anterior pelvic tilt increase) showed a lowered change for CJT-1 compared to SMB-1 (Figure 6a and Table S1).

Elongation was significantly higher during the initial anterior pelvic tilt degrees compared to the final ones both for CJT-1 and for SMB-1 see Figure 6b and Tables S1 and S2.

At the distal hamstring region, significant changes were observed for BF-2, SMB-2 and ST-3 at 0–25° compared to initial anterior pelvic tilt (0–5°) (see Figure 7a and Table S4). Nonsignificant differences were found between muscles (Tables S3 and S5). ST-3 elongation for every five anterior pelvic tilt change degrees differed significantly in each section (see Figure 7b and Table S4). However, the length change for BF-2 and SMB-2 was significantly higher for the final anterior pelvic tilt angles compared to the initial ones (see Figure 7b and

FIGURE 3 Set-up of the specimen with markers and cameras in position. The markers were placed in nine different locations within the hamstring muscles, and the two cameras measured all changes in the position of the markers at each anterior pelvic tilt angle.

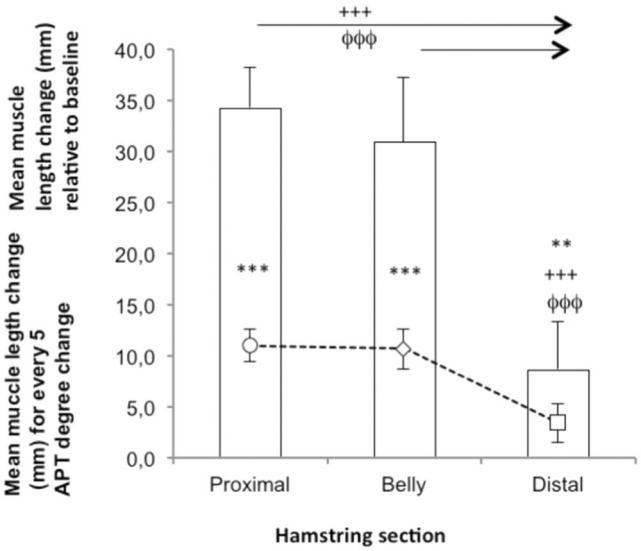

FIGURE 5 Mean length change compared to origin and SD at proximal • (CJT1, CJT2, SMB1), belly ♦ (BF1, ST1 and ST2) and distal ■ (BF2, SMB2 and ST3) hamstring region. +, significant differences related to proximal hamstring location; ϕ, significant differences compared to medial hamstring location; *significant differences compared to origin hip position. Two symbols $p < 0.01$ and three symbols $p < 0.001$. BF, biceps femoris; CJT, conjoint tendon; SMB, semimembranosus; ST, semitendinosus.

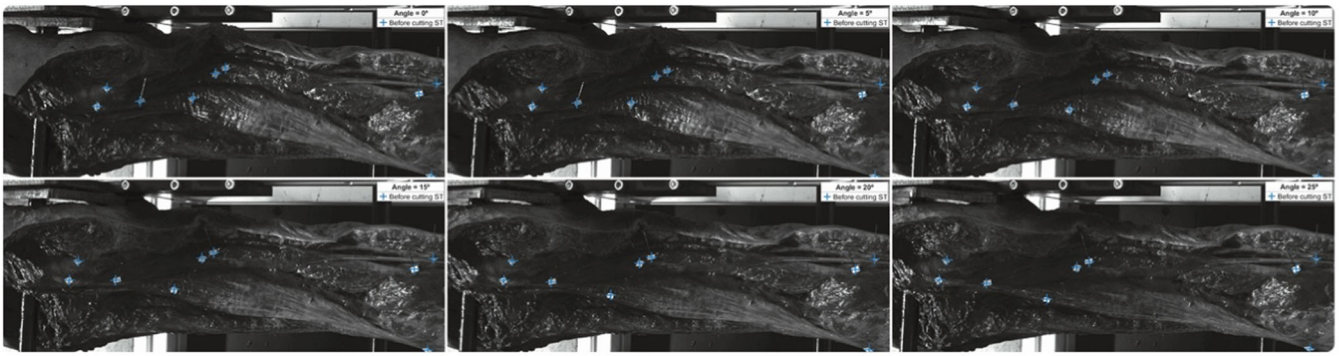

FIGURE 4 Changes in position of the nine markers at each anterior pelvic tilt angle. Notice the progressive lengthening of the muscles with increasing anterior pelvic tilt angle.

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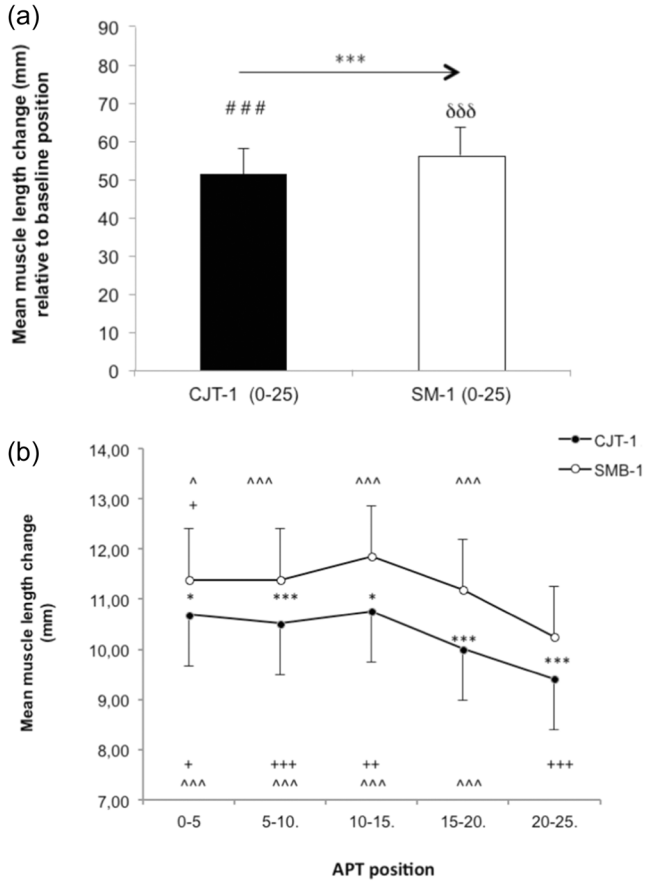

FIGURE 6 Proximal hamstring changes (CJT-1 and SMB-1) relative APT angle. (a) Mean length change relative to origin and SD. (b) Mean length change and SD for every 5° anterior pelvic tilt change. Black coloured bars or dots CJT-1, white coloured SMB-1. Mean muscle length change for CJT-1 and SMB-at 25 was significantly different compared to baseline ♯ = $p < 0.000$ and for SMB-1 δ = $p < 0.000$. For further information see also Table S1. CJT-1 elongation was significantly different compared to SMB-1. CJT-1 and SMB-1 elongation were significantly different compared to elongation at 15–20°+ and/or 20–25°^ anterior pelvic tilt degrees. One symbol $p < 0.05$, two $p < 0.01$ and three $p \leq 0.001$ (see Table S2). CJT, conjoint tendon; SMB, semimembranosus.

Table S4). No significant differences were found between BF-2 compared to SMB-2 or ST-3 or SMB-2 compared to ST-3 relative to the baseline (see Table S4) or for every 5° anterior pelvic tilt increase (see Table S5).

DISCUSSION

The main finding of the study is that an increase in anterior pelvic tilt caused a significant elongation in all three hamstring muscles (SMB, ST and long head BF). The increase in elongation was greater in the proximal and middle sections compared to the distal sections.

To determine the effect of pelvis biomechanics on hamstring elongation we developed an in-vitro study to systematically evaluate the effect of differential anterior pelvic tilt positions concomitant with precise measurement of the deformation patterns of the different hamstring muscle complex. Musculoskeletal computer simulation models have been widely used in the literature to estimate hamstring muscle–tendon length with the specific goal to improve clinical decision-making and surgical outcomes in children with cerebral palsy or in sports medicine to describe the mechanical behaviour of the posterior thigh muscles during hamstring strain injury main injury mechanism (sprint) [24–26]. These models based their estimations on static parameters derived from human cadavers, such as muscle fibre length, pennation angles, optimum fibre length, physiologic cross-sectional area, origin to insertion length and joint angles; when correlated with arm-based coordinates (to determine muscle length change) directly calculated from human specimens, these models perform with accuracy [20]. Arm-based estimations are directly calculated by moving the femur over the pelvis to calculate hip flexion angles similar to those found during walking [20]. This method of estimation represents a simpler perspective to model hip flexion since the estimate can also be obtained the other way around, moving the pelvis around the femur.

Our study may provide novel methods and results to improve current musculoskeletal models, where the hip joint angle is defined as the angle between the pelvis and femur, and pelvic tilt is not considered in its appropriate magnitude regarding hamstring lengthening. Our results showed a significant increase in tissue elongation in each of the individual muscles and regions composing the hamstring muscle complex with any increase in anterior pelvic tilt. It would also be useful to compare our results with the most commonly used hamstring musculoskeletal models [24–26].

As an example, taking as a reference a 1.70–1.80-m subject, the current models estimate a change of 0.5 cm and −0.2 for every 5° of hip and knee flexion. If our in vitro data are applied, 5° of anterior pelvic tilt would result in more than 1 cm elongation at the proximal level and around 0.4 cm at the distal level. Consequently, advanced musculoskeletal models and simulation techniques [24–26] may be underestimating the hamstring muscle-tendon complex lengthening during high-speed running, since those models do not consider the pelvis as an independent segment and do not compute the pelvis with respect to the femur.

During the late-swing phase of maximal speed sprinting, the BF faces an elongation at the proximal level (±4 cm), and the strain experienced at this level may be directly influenced by the anterior pelvic tilt. Thus, the current results indicate a mechanistic basis to expect that a decrease in anterior pelvic tilt would reduce the tensile strength of the proximal region (highest strain location) of the most injured muscle (i.e., BF) during high-speed running [16]. A recent evaluation of a 6-week multimodal intervention combining lumbopelvic control exercises with a running

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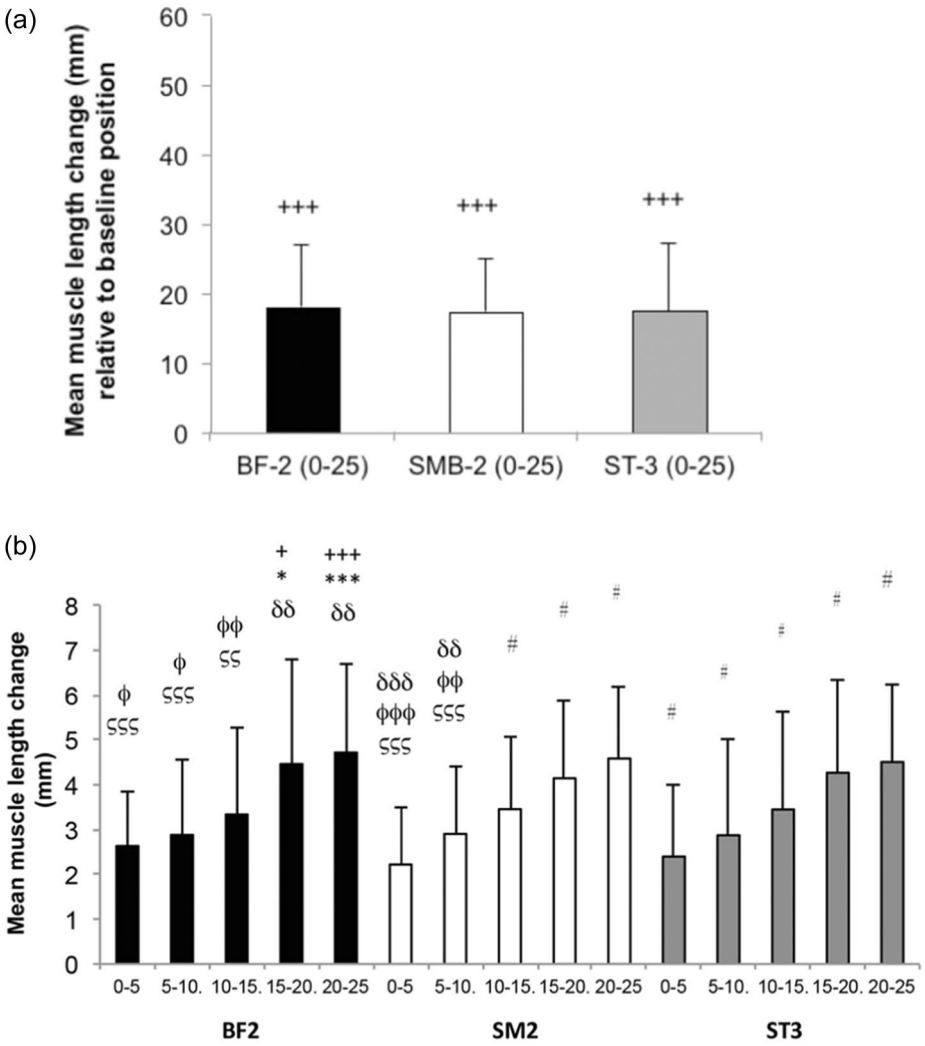

FIGURE 7 Distal hamstring changes (BF-2 and SMB-2 and ST-3) relative APT angle. (a) Mean length change relative to origin and SD. (b) Mean length change and SD for every 5° anterior pelvic tilt change. Black coloured bars BF-2, white coloured SMB-2 and grey coloured = ST-3. Mean accumulated length increase relative to all other angles within the same muscle $^{\#}p < 0.01$, significantly different compared to the elongation at 0–5° = +, 5–10° = *, 10–15° = δ, 15–20° = ϕ and 20–25° = ς. See Tables S4 and S5. BF, biceps femoris; SMB, semimembranosus; ST, semitendinosus.

technique programme revealed a significant decrease in anterior pelvic tilt (5°) during sprinting at the late-swing phase [23]. This change would be equivalent to reducing about 1 cm from the 4 cm of maximum elongation reached by the proximal region of the BF muscle thereby reducing peak hamstring strain during sprinting.

Assuming that disproportionate strain is the greatest determinant of hamstring muscle injury [13], the current study places the pelvis as a key factor to consider when designing intervention exercises aiming to decrease strain on the hamstring muscle complex. This contradicts the predominance of hip-dominant (mainly femur-focused) and knee-dominant classification for strengthening exercises previously proposed to prevent hamstring injury [27, 28]. In addition, the pelvis will play a fundamental role as a strain regulator, impacting rehabilitation management of pathologies that specifically affect the proximal region of the hamstring muscles, such as tendinopathies or avulsions.

Comparison among hamstring muscles and locations

In the proximal region, we observed a more significant increase in the elongation of the SMB free tendon compared with respect to the conjoint tendon. These findings support previous studies using elastography which observed a greater rigidity of the ST and BF compared to the SMB at any anterior pelvic tilt position and at hip and knee extreme arrangement [21]. Overall, these results are similar to the findings of Askling et al., where acute hamstring strain injuries affecting the proximal free tendon of the SMB (2.5 cm of ischial

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tuberosity) were associated with movements involving extreme joint angles (sagittal split, side split, stretching and high kick) in different studies and athletes (footballers and dancers) [29, 30]. Morphological factors (free tendon length, medial thickness) and its spiral course (lateral to medial) could predispose the SMB to a significant stretch [31], particularly during extreme ranges of movement when the muscle is extended across both the knee and hip joints and especially in the presence of anterior pelvic tilt. In summary, analysis of injury mechanism(s) and rehabilitation strategies for injuries that affect the proximal free tendon of the SMB should specifically consider dynamic pelvis biomechanics.

The analysis of the distal region of the hamstring muscle complex did not reveal differences related to the increased elongation of the hamstring muscles as anterior pelvic tilt increased. Recent clinical evidence indicates that injuries located at the distal musculotendinous junction of the BF, represent an emerging problem with a high recurrence rate (54%) among hamstring strain injury types [32]. Increased injury rates and recurrence in this region may stem from the combination of a different nerve supply and the 45° insertion angle of both BF heads on the BF distal tendon, resulting in different angles of pull [32, 33]. Our findings revealed a higher percentage of distal tendon lengthening of the BF-2 with higher versus lower degrees of anterior pelvic tilt (see Table S4). Notably, exercises such as Nordic and stiff-leg deadlifts in their most extreme ranges (where they share a large anterior pelvic tilt) result in greater activation of the distal region of the BF long-head muscle [34, 35]. This phenomenon could explain the difference between the region and the most affected hamstring muscle during the sprint injury mechanism (proximal BF long head) and the actions associated with this type of distal injury such as scrums, rucks and mauls characteristic of rugby [36]. Also evident as the support leg during kicking [37], where a greater degree of anteversion of the pelvis, hip flexion, and knee (hyper)extension seem to impose more stress on the distal long-head tendon of the BF. Therefore, the pelvis represents an important segment to consider in the management of the increasingly common hamstring muscle injuries that affect the distal region of the long head of the BF.

Of note, one limitation of this study is the small number of specimens used, although the results were very consistent among all seven specimens. Another inherent limitation of the use of human cadavers is the impossibility of knowing the impact of muscle contraction and fatigue on muscle strain. Furthermore, and although the aim of the study was to know the isolated effect of pelvic movement on a fixed femur and tibia on the hamstring strain, it would be interesting for future projects to know the interaction of the pelvis in conjunction with dynamic hip and knee free motion as it happens in sports actions such as the sprint.

CONCLUSION

This study showed that an increase in anterior pelvic tilt results in a significant nonuniform increase in tissue elongation in all regions of the three hamstring muscles (SMB, ST, BF long head). This increase is greater in the proximal compared to the distal region. These results will have an impact on injury rehabilitation and prevention processes of hamstring injuries such as avulsions, tendinopathies, or strain injuries, as well as optimize future musculoskeletal models and avoid potential underestimation of the hamstring muscle-tendon complex lengthening during high-speed running.

AUTHOR CONTRIBUTIONS

Conceived and designed the experiments: Jurdan Mendiguchia, Miki Dalmau-Pastor. *Performed experiments*: Jurdan Mendiguchia, Miki Dalmau-Pastor. *Analyzed data*: Jurdan Mendiguchia, Miki Dalmau-Pastor, Mirian Aranzazu Garrues. *Interpreted results of research*: Jurdan Mendiguchia, Miki Dalmau-Pastor, Mirian Aranzazu Garrues, Ernest Schilders, Gregory D. Myer. *Drafted manuscript and prepared tables/figures*: Jurdan Mendiguchia, Miki Dalmau-Pastor, Mirian Aranzazu Garrues. *Edited, critically revised paper and approved final version of manuscript*: Jurdan Mendiguchia, Miki Dalmau-Pastor, Mirian Aranzazu Garrues, Ernest Schilders, Gregory D. Myer.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

ETHICS STATEMENT

This research was carried out in the dissection room of the University of Barcelona, following ethics permission under IRB number 00003099.

ORCID

Jurdan Mendiguchia http://orcid.org/0000-0001-5833-2890
Mirian Aranzazu Garrues http://orcid.org/0000-0002-6446-4993
Ernest Schilders http://orcid.org/0000-0002-3318-944X
Gregory D. Myer http://orcid.org/0000-0002-9983-8422
Miki Dalmau-Pastor http://orcid.org/0000-0001-6043-698X

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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